of energy.

30 A radioactive source in the laboratory has a half-life of 10 days. The count rate was measured to be 100 Bq initially. 20 days later, the count rate was found to be 34 Bq. What is the count rate in the laboratory without the source?

A 9 **B** 12 **C** 17 **D** 22

Ans: B

$(C - C_{\text{background}}) = \frac{1}{4}(C_0 - C_{\text{background}})$

$(34 - C_{\text{background}}) = \frac{1}{4}(100 - C_{\text{background}})$

$C_{\text{background}} = 12$ days

End of Paper